

34. Morphological Top Hat

Aim

To perform morphological Top Hat operation on an image using OpenCV.

Procedure

1. Read the image.
2. Convert it to grayscale.
3. Create a kernel.
4. Apply Top Hat transformation.
5. Display the output.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.ones((5, 5), np.uint8)

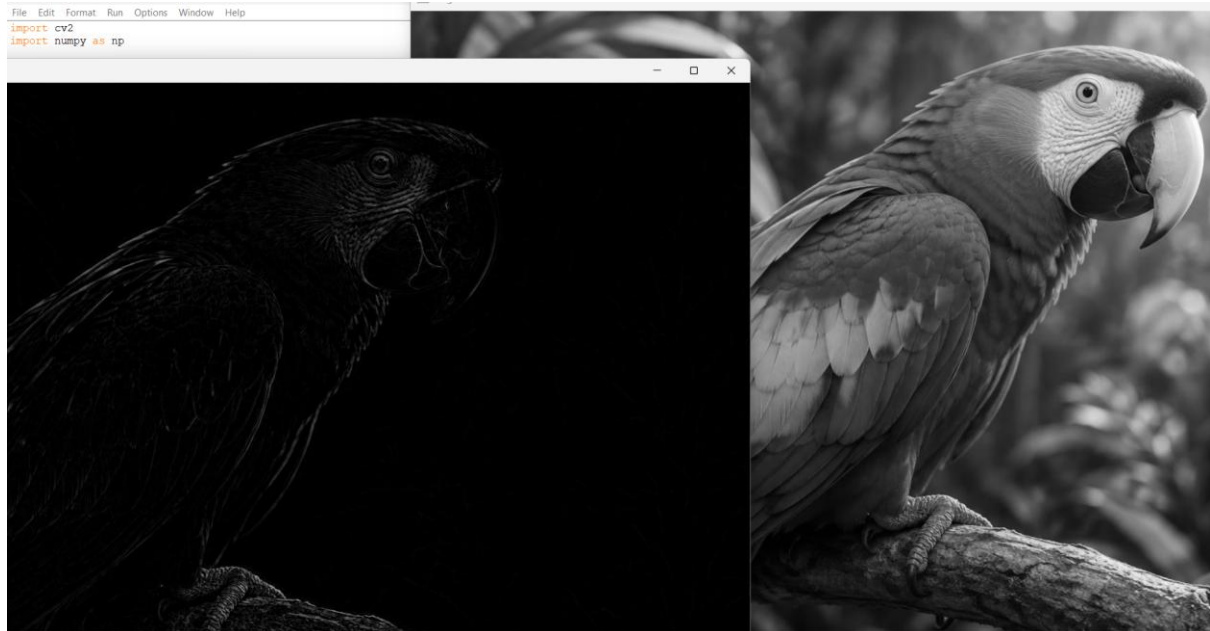
top_hat = cv2.morphologyEx(
    gray,
    cv2.MORPH_TOPHAT,
    kernel
)

cv2.imshow("Original", gray)
cv2.imshow("Top Hat", top_hat)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Output



Result

The Top Hat morphological transformation was successfully performed on the given image.